EXPERIMENT NO. 11 (Group B)

Aim: Data Aggregation Problem Statement: Analyzing Sales Performance by Region in a Retail Company Dataset: "Retail_Sales_Data.csv" Description: The dataset contains information about sales transactions in a retail company. It includes attributes such as transaction date, product category, quantity sold, and sales amount. The goal is to perform data aggregation to analyze the sales performance by region and identify the top-performing regions.

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Outputs:

Step 1 — Import and Load Dataset

In [2]:
```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("customer_shopping_data.csv")

print("Dataset loaded successfully!")
print("Shape:", df.shape)

display(df.head())
```

```
Dataset loaded successfully!
Shape: (99457, 10)
```

	invoice_no	customer_id	gender	age	category	quantity	price	payment_method	inv
0	I138884	C241288	Female	28	Clothing	5	1500.40	Credit Card	
1	I317333	C111565	Male	21	Shoes	3	1800.51	Debit Card	1
2	I127801	C266599	Male	20	Clothing	1	300.08	Cash	
3	I173702	C988172	Female	66	Shoes	5	3000.85	Credit Card	1
4	I337046	C189076	Female	53	Books	4	60.60	Cash	2

Step 2 — Explore and Clean Data

In [3]:
```
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")

for col in df.select_dtypes(include="object").columns:
    df[col] = df[col].astype(str).str.strip()

df["quantity"] = pd.to_numeric(df["quantity"], errors="coerce")
df["price"] = pd.to_numeric(df["price"], errors="coerce")

df = df.dropna(
```

```
    subset=["shopping_mall", "category", "quantity", "price"]
)

print("Columns:")
print(df.columns.tolist())

print("\nMissing Values:")
print(df.isnull().sum())
```

```
Columns:
['invoice_no', 'customer_id', 'gender', 'age', 'category', 'quantity', 'price', 'pay
ment_method', 'invoice_date', 'shopping_mall']

Missing Values:
invoice_no        0
customer_id       0
gender            0
age               0
category          0
quantity          0
price             0
payment_method    0
invoice_date      0
shopping_mall     0
dtype: int64
```

Step 3 — Total Sales by Mall

In [4]:
```
df["sales_amount"] = df["price"] * df["quantity"]

branch_sales = (
    df.groupby("shopping_mall")["sales_amount"]
    .sum()
    .sort_values(ascending=False)
)

print("Total Sales by Shopping Mall:")
print(branch_sales)

print("\nTop Performing Mall:")
print(branch_sales.head(1))
```

```
Total Sales by Shopping Mall:
shopping_mall
Mall of Istanbul     50872481.68
Kanyon               50554231.10
Metrocity            37302787.33
Metropol AVM         25379913.19
Istinye Park         24618827.68
Zorlu Center         12901053.82
Cevahir AVM          12645138.20
Viaport Outlet       12521339.72
Emaar Square Mall    12406100.29
Forum Istanbul       12303921.24
Name: sales_amount, dtype: float64

Top Performing Mall:
shopping_mall
Mall of Istanbul    50872481.68
Name: sales_amount, dtype: float64
```

Step 4 — Sales Visualization

In [5]:

```
category_sales = (
    df.groupby("category")["sales_amount"]
    .sum()
    .sort_values(ascending=False)
)

print("Sales by Category:")
print(category_sales)

plt.figure(figsize=(10, 5))
plt.bar(branch_sales.index, branch_sales.values)

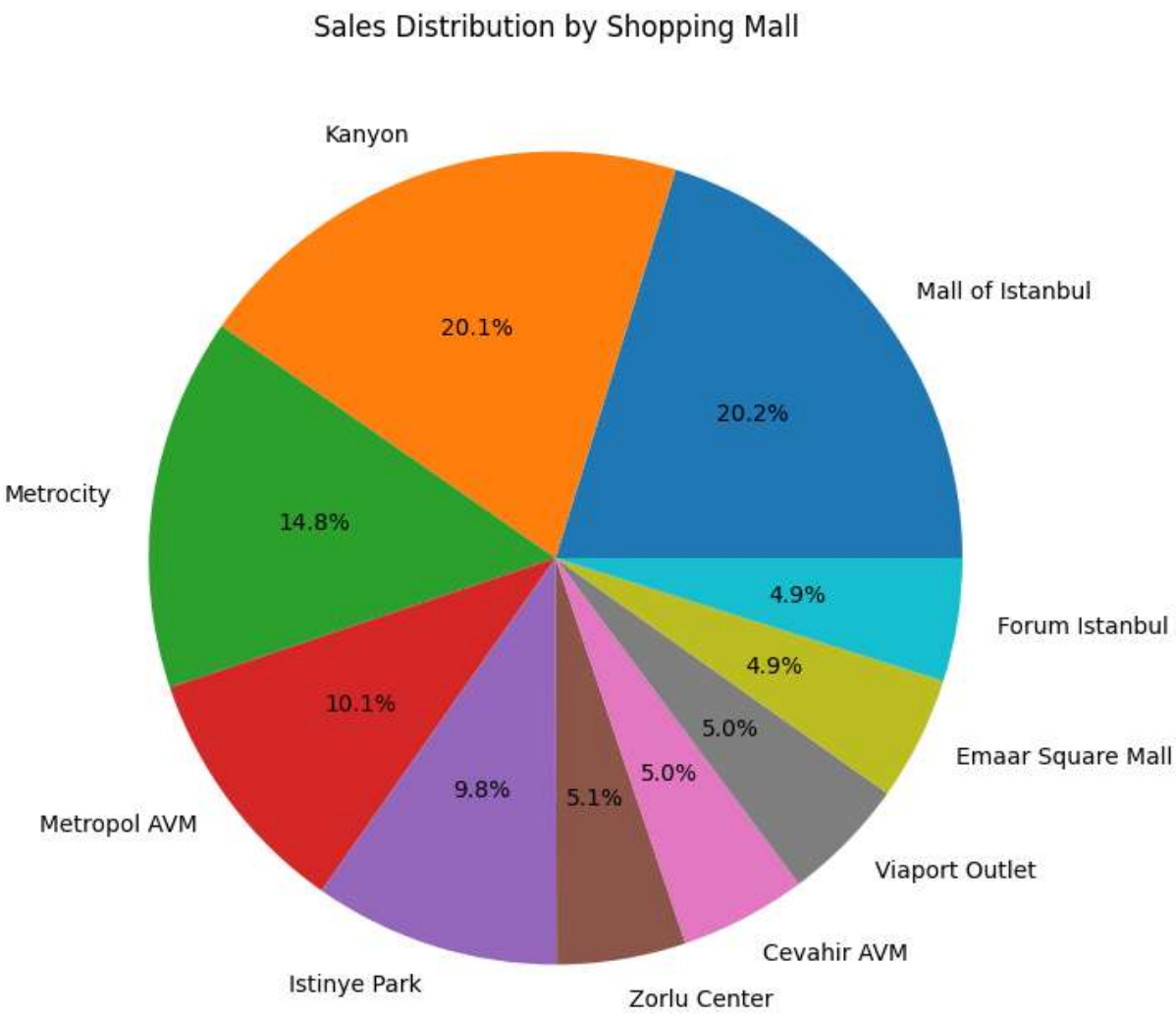
plt.xlabel("Shopping Mall")
plt.ylabel("Total Sales")
plt.title("Sales Distribution by Shopping Mall")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

plt.figure(figsize=(8, 8))
plt.pie(
    branch_sales.values,
    labels=branch_sales.index,
    autopct="%1.1f%%"
)

plt.title("Sales Distribution by Shopping Mall")
plt.show()
```

```
Sales by Category:
category
Clothing           1.139968e+08
Shoes              6.655345e+07
Technology         5.786235e+07
Cosmetics          6.792863e+06
Toys               3.980426e+06
Food & Beverage    8.495351e+05
Books              8.345529e+05
Souvenir           6.358247e+05
Name: sales_amount, dtype: float64
```





Step 5 — Mall + Category Aggregation

In [7]:
```
combined_pivot = df.pivot_table(
    index="shopping_mall",
    columns="category",
    values="sales_amount",
    aggfunc="sum",
    fill_value=0
)

print("Sales by Mall and Category:")
display(combined_pivot)

combined_pivot.plot(
    kind="bar",
    figsize=(14, 6)
)

plt.xlabel("Shopping Mall")
plt.ylabel("Total Sales")
plt.title("Sales by Shopping Mall and Product Category")
plt.xticks(rotation=45)
```

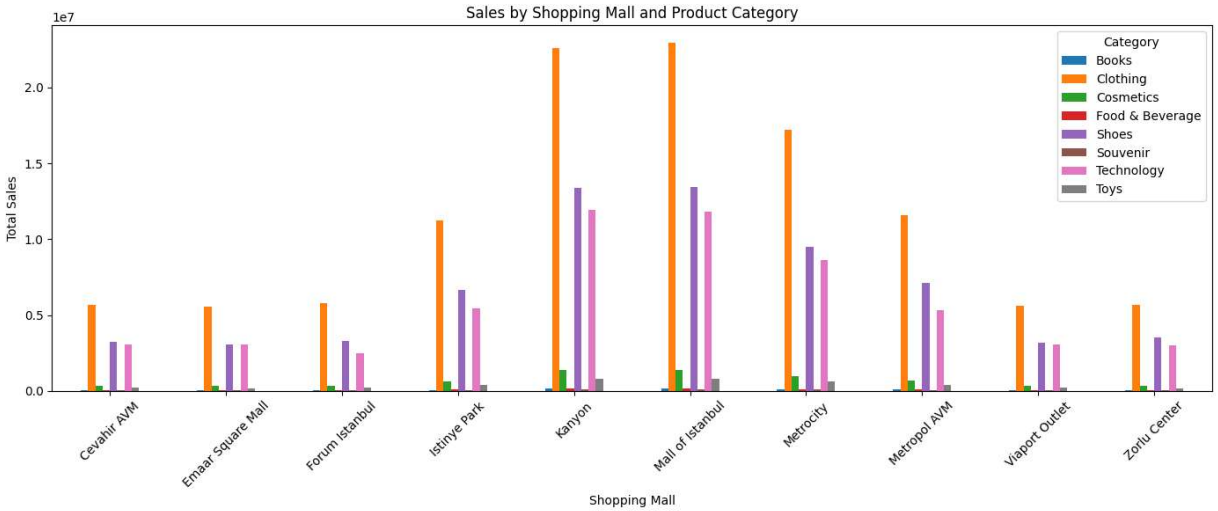
```
plt.legend(title="Category")
plt.tight_layout()
plt.show()

print("\nTop 5 Mall-Category Sales:")
print(
    df.groupby(["shopping_mall", "category"])["sales_amount"]
    .sum()
    .sort_values(ascending=False)
    .head(5)
)


print("Aditya Rade  Roll.No:BEAD23167")
```

```
Sales by Mall and Category:
```

category	Books	Clothing	Cosmetics	Food & Beverage	Shoes	Souvenir	Techn
shopping_mall							
Cevahir AVM	44541.00	5706321.28	321214.00	44010.45	3243918.85	29723.82	305
Emaar Square Mall	41995.80	5590490.40	338941.76	40610.95	3089675.16	30943.74	309
Forum Istanbul	42056.40	5792444.24	353172.76	39162.24	3327942.65	32879.19	251
Istinye Park	76083.30	11253900.24	655357.88	85918.44	6641481.22	68925.48	543
Kanyon	163029.15	22609527.60	1369550.78	166497.05	13383190.83	127399.53	1194
Mall of Istanbul	172240.35	22947417.68	1367517.78	171177.90	13467814.80	127540.29	1182
Metrocity	125911.65	17226692.56	991860.04	129902.74	9519296.37	94227.09	860
Metropol AVM	83718.90	11568084.00	680770.38	88638.04	7149825.21	67869.78	532
Viaport Outlet	39632.40	5604594.16	347439.70	41662.18	3194704.91	27319.17	306
Zorlu Center	45343.95	5697318.88	367037.82	41955.06	3535601.47	28996.56	298




```
Top 5 Mall-Category Sales:
shopping_mall     category
Mall of Istanbul  Clothing    22947417.68
Kanyon            Clothing    22609527.60
Metrocity         Clothing    17226692.56
Mall of Istanbul  Shoes       13467814.80
Kanyon            Shoes       13383190.83
Name: sales_amount, dtype: float64
Aditya Rade  Roll.No:BEAD23167
```